HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL & COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS & STATISTICS

F78AA1
ACTUARIAL AND FINANCIAL MATHEMATICS A

9 December 2009
Time allowed 2 hours

Please answer **ALL 5** questions.

Total marks: 60

Only approved calculators may be used.

The examination will be marked anonymously. Please ensure you have entered your name and details as instructed and then seal the flap in order to conceal these.

1.

(i) At an effective rate of interest of 4% per half year effective calculate:-

(a) ${}_5|\ddot{a}_{\overline{20}|}$

(b) $s^{(4)}_{\overline{15}|}$

[4 marks]

(ii) Show that $a_{\overline{15}|} = v^{15} s_{\overline{15}|}$

[3 marks]

[**Total 7 marks**]

2. An investor is considering the purchase of an annuity, payable annually in arrear for 20 years. The first payment is £300.

(i) Using a rate of interest of 10% per annum effective, calculate the present value of the annuity when:

(a) the payments remain level over the term

(b) the payments increase at a rate of 12% per annum compound

(c) the payments increase at a rate of 12% per annum simple

[9 marks]

(ii) Explain why the answer in (i)(b) is higher than in (i)(c)

[2 marks]

(iii) Another investor invests £1,000 for 2 years but earns interest at an effective rate of discount of 6% per annum. He invests no further money over the next 3 years, but his existing savings earn interest at a force of interest of 7% per annum during this period (ie for $t = 2$ to $t = 5$). What is the total value of his savings after 5 years?

[4 marks]

[**Total 15 marks**]

[**Please turn over**]

3. On 15 August 2005 a financial institution issued a bond with term to redemption of 15 years and a coupon of 5% per annum. Coupons are paid yearly in arrear on 15 August and the bond will be redeemed at 120% nominal.

An investor purchased £20,000 nominal of this bond on 16 August 2008 for £22,000. He pays income tax at a rate of 20% and capital gains tax at a rate of 40%.

(a) Using the formulae for the upper and lower bounds of a fixed interest security and linear interpolation (or otherwise), calculate his net effective yield if he holds the bond to maturity.

[6 marks]

Coupons and redemption proceeds were linked to the inflation index below with a 2-month time lag. In August 2009 the recorded values of the inflation index were:-

Date	**Index**
Base	192.2
August 2005	194.1
June 2008	216.8
August 2008	217.8
September 2008	218.4
December 2008	212.9
February 2009	213.1
March 2009	211.3
June 2009	213.4

(b) Show that the first coupon payment the investor will receive (before the deduction of income tax) will be £1,110.30. [2 marks]

(c) Assuming inflation of 5% per annum from June 2009, what will the redemption proceeds be? [3 marks]

(d) Another investor, who does not pay any income or capital gains tax, bought £20,000 nominal of this bond on 16 February 2009. Assuming inflation of 5% per annum from 16 August 2009, what price did the investor pay if he achieved an effective rate of return of 10% per annum? [5 marks]

[**Total 16 marks**]

[**Please turn over**]

4. An individual takes out a 20-year bank loan of £200,000 to purchase a house. He agrees to pay only the interest payments, monthly in arrear and no capital will be repaid until the end of the term (ie after 20 years).

(a) At an effective interest rate of 6% per annum effective, what is the monthly repayment? How much does the investor pay in interest over the entire loan? [2 marks]

(b) The individual plans to pay off the loan after 10 years by making monthly payments into a separate account. If he earns interest in this account at 9% per annum effective, how much does he need to save each month in order to have enough money in his savings account to repay the entire loan after 10 years?
[3 marks]

(c) Over the first ten years, he actually earned only 5% per annum effective on his savings. Show that his savings were worth £154,363 after 10 years and hence that the outstanding loan would be £37,272 if he used all his savings to reduce the loan.
[3 marks]

(d) At this point he decided to take out a new loan of £37,272 for 15 years at 7% per annum effective with level instalments of interest and capital paid monthly in arrear. How much is the level repayment? [2 marks]

(e) Show that the individual has paid interest at an effective rate of 6.55% per annum over the entire term of the loan.
[4 marks]

[**Total 14 marks**]

[**Please turn over**]

5. A loan of nominal amount £100,000 is to be issued bearing coupons payable half-yearly in arrear at a rate of 8% per annum. Capital is to be redeemed at 110% on a coupon date 15 years after the date of issue. An investor who is liable to income tax at 25% and capital gains tax at 35% wishes to purchase the entire loan at the date of issue.

(a) Show that the price he should pay to ensure a net effective yield of 5% per annum is £115,959. [4 marks]

(b) If the term were 20 years, what price should he pay? [2 marks]

(c) If the borrower can choose to repay the loan at any time from 15 to 20 years, and the market rate of interest is 5% per annum effective, when would he choose to repay the loan and why? [2 marks]

[**Total 8 marks**]

END OF PAPER

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